

# LECTURE NOTES: CHAPTER 2

SECTIONS 2.1, 2.2, 2.3, 2.4, 2.5, 2.6

FALL, 2019

---

## CONTENTS

|   |                                               |    |
|---|-----------------------------------------------|----|
| 1 | Introduction                                  | 3  |
| 2 | Section 2.1 - The Idea of Limits              | 4  |
| 3 | Section 2.2 - Definitions of Limits           | 9  |
| 4 | Section 2.3 - Techniques for Computing Limits | 11 |
| 5 | Section 2.4 - Infinite Limits                 | 16 |
| 6 | Section 2.5 - Limits at Infinity              | 19 |
| 7 | Section 2.6 - Continuity                      | 24 |



## INTRODUCTION

In many situations in an average person's day-to-day life, one will encounter things that are related to each other in some fashion (in fact, in a gratuitously nit-picky sense, one could argue that *absolutely everything* we encounters satisfies this criteria due to the laws of physics). When things are related in some fashion, we can generally figure out some way to translate this relation into symbols in some manner. Usually, this involves representing the things by their own symbols, the way they interact together by some other symbols, and the relationship itself by another symbol (which we normally call the equals sign).

For example, let's say I get on the interstate and start driving, and let's say I notice that I see a mile marker every minute or so. I think about this for a bit, then realize that this means I've traveled one mile every minute. If I wanted to represent this with symbols, I could decide to denote the number of miles I've traveled from since I got on the interstate by the symbol "x" and the number of minutes I've driven by the symbol "y". So, since I've traveled about one mile every minute, I could write something like

$$y = x$$

as a way to say that the number of minutes I've driven is the same as the number of miles I've traveled.

But now, what if I wanted to have "x" represent how far away (in miles) I am from my house instead of how far away I am from the entrance to the interstate? Well, if my house is about 5 miles from the interstate, then I would just have to add five miles to the distance I've traveled. So, I would write something like

$$y = x + 5$$

instead. But now, even though I have two different ways of writing two different things in symbols, I notice that *part of what I'm describing is the same*. Namely, if I drive for one more minute, I'm one mile further than I was a minute ago – no matter what number of miles I had previously traveled before the minute of driving had passed. This is one of the key ideas that calculus deals with, and we'll explore more nuanced variations on this theme throughout the course of this class.

However, before we get to that, we need to figure out some *basic properties* of these types of symbolic representations of relationships between quantities. In this chapter, we'll build a tool that will allow us to do this – the *limit*, as we call it – and we'll discuss the first of these properties – *continuity*.

## SECTION 2.1 - THE IDEA OF LIMITS

Limits are a powerful tool in mathematics that underlie both of the key concepts in calculus – derivation and integration. We’ll talk briefly in lecture about the general idea of what exactly a limit is, but we’ll primarily focus on motivating the idea of a limit in this section. To do so, we’ll look at two real-world problems almost all of us have encountered before: calculating average velocity and calculating instantaneous velocity. Then, we’ll look at two abstract problems which turn out to be, essentially, the same as the real-world problems: calculating the slope of a secant line and calculating the slope of a tangent line.

**Definition 1** (Average Velocity).

Let  $f(t)$  be a function describing the position of some object at time  $t$ . Then, given a starting time  $t_0$  and a number  $h \neq 0$ , consider the interval  $[t_0, t_0 + h]$  (that is, all of the times  $t$  such that  $t_0 \leq t \leq t_0 + h$ ). We say the *average velocity* of the object over the interval  $[t_0, t_0 + h]$  is the quantity

$$v_{avg} = \frac{f(t_0 + h) - f(t_0)}{(t_0 + h) - t_0} = \frac{f(t_0 + h) - f(t_0)}{h}.$$

That is to say, average velocity is the *change in position* divided by the *change in time*.

**Example 1.** Suppose you’re taking a long road trip and you stop for gas. When you stop, you look at a map and notice that you’ve driven 120 miles since you last stopped for gas an hour and a half ago. What was your *average velocity* during the last hour and a half?

**Example 2.** Suppose you throw a rock straight upwards with a speed of  $96\frac{ft}{s}$ . Neglecting air resistance, a well-known formula from physics states that the position of the rock after  $t$  seconds is given by the function

$$s(t) = -16t^2 + 96t.$$

What was the *average velocity* of the rock between  $t = 1$  s and  $t = 3$  s? What about between  $t = 1$  s and  $t = 2$  s? Between  $t = 1$  s and  $t = 1.1$  s?

**Definition 2** (The Slope of a Secant Line).

Let  $f(x)$  be a function. Then, given  $x$ -values  $x_0$  and  $x_1$  with  $x_0 < x_1$ , we know that the points  $(x_0, f(x_0))$  and  $(x_1, f(x_1))$  lie on the *graph* of the function  $f$ . The line which passes through both of these points is called a *secant line*, and has a slope of

$$m_{sec} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}.$$

**Example 3.** Suppose you have the function

$$f(x) = x^2 + x + 1.$$

What is the slope of the *secant line* passing through the points  $(1, f(1))$  and  $(2, f(2))$ ?

**Example 4.** Suppose you have the function

$$f(x) = 2^x + \frac{1}{x}.$$

What is the slope of the *secant line* passing through the points  $(1, f(1))$  and  $(2, f(2))$ ?

**Definition 3** (Instantaneous Velocity).

Let  $f(t)$  be a function describing the position of some object at time  $t$ . Then, given a time  $t_0$ , we say the *instantaneous velocity* of the object is the quantity

$$v_{inst} = \lim_{h \rightarrow 0} \frac{f(t_0 + h) - f(t_0)}{h}.$$

That is to say, average velocity is value approached by the *change in position* divided by the *change in time* as time becomes *arbitrarily close to zero*.

**Example 5.** Consider the rock thrown in Example 2 and recall that the position of the rock after  $t$  seconds is given by the function

$$s(t) = -16t^2 + 96t.$$

Estimate the *instantaneous velocity* of the rock at  $t = 1$  s by calculating the average velocities over the intervals given below:

1.  $[1, 1.01]$ :

2.  $[1, 1.001]$ :

3.  $[1, 1.0001]$ :

**Definition 4** (The Slope of a Tangent Line).

Let  $f(x)$  be a function. Then, given an  $x$ -value  $x_0$ , we say the slope of the *tangent line* of  $f$  at  $x_0$  is the quantity

$$m_{tan} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

That is to say, the slope of the tangent line is the value approached by the slopes of the secant lines between  $(x_0, f(x_0))$  and  $(x_0 + h, f(x_0 + h))$  as  $h$  becomes arbitrarily small.

**Example 6.** Consider the function

$$f(x) = x^2.$$

Estimate the *slope of the tangent line* at the point  $x = 1$  by calculating the slope of the secant line between the points given below:

1.  $(1, f(1))$  and  $(1.1, f(1.1))$ :

2.  $(1, f(1))$  and  $(1.01, f(1.01))$ :

3.  $(1, f(1))$  and  $(1.001, f(1.001))$ :



## SECTION 2.2 - DEFINITIONS OF LIMITS

Now that we've motivated the idea of limits with tangent lines and instantaneous velocities, let's define what a limit is. This isn't the rigorous definition of a limit (which we'll touch on in section 3.1), but it will serve our needs for the time being.

**Definition 5** (One-Sided Limits).

Let  $f(x)$  be a function and let  $c$  be some fixed number such that  $f(x)$  is defined for all  $x$  near  $c$  (except, possibly, at  $c$  itself). Then, for some number  $L$ , we write

$\lim_{x \rightarrow c^-} f(x) = L$  if the value of  $f(x)$  approaches  $L$  as  $x$  approaches  $c$  from the left;

$\lim_{x \rightarrow c^+} f(x) = L$  if the value of  $f(x)$  approaches  $L$  as  $x$  approaches  $c$  from the right.

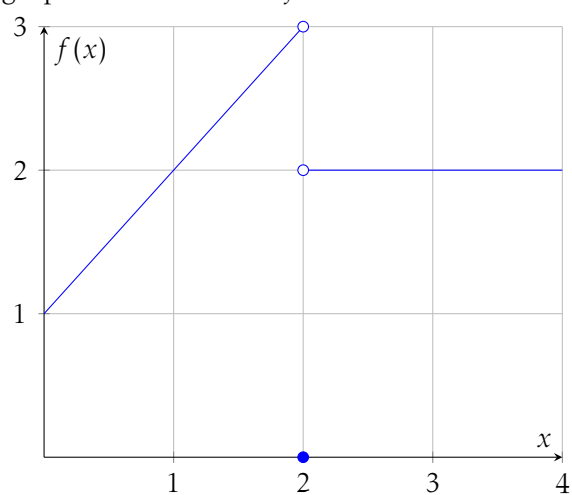
**Definition 6** (Limits).

Let  $f(x)$  be a function and let  $c$  be some fixed number such that  $f(x)$  is defined for all  $x$  near  $c$  (except, possibly, at  $c$  itself). Then, for some number  $L$ , we write

$$\lim_{x \rightarrow c} f(x) = L \text{ if } \lim_{x \rightarrow c^-} f(x) = L \text{ and } \lim_{x \rightarrow c^+} f(x) = L.$$

**Example 7.** If  $f(x) = \begin{cases} x + 2 & \text{if } x \leq 0 \\ 2 - x & \text{if } x > 0 \end{cases}$ , find  $\lim_{x \rightarrow 0} f(x)$ .

**Example 8.** Use the graph of the function  $f$  shown to estimate the indicated limits.



1.  $\lim_{x \rightarrow 2^-} f(x)$

2.  $\lim_{x \rightarrow 2^+} f(x)$

3.  $\lim_{x \rightarrow 2} f(x)$

## SECTION 2.3 - TECHNIQUES FOR COMPUTING LIMITS

While we have an idea of what limits are, we don't have many ways of actually computing them so far. In this section, we'll learn a variety of *properties of limits* which will allow us to compute the limit of a function in many more scenarios.

**Theorem 1** (Properties of Limits).

Let  $f$  and  $g$  be functions, and assume  $\lim_{x \rightarrow c} f(x)$  and  $\lim_{x \rightarrow c} g(x)$  exist. Let  $a$  and  $b$  be real numbers, and let  $m$  and  $n$  be positive integers. Then, the following properties hold:

1. **Linear Functions:**  $\lim_{x \rightarrow c} (ax + b) = mc + b$
2. **Sum:**  $\lim_{x \rightarrow c} (f(x) + g(x)) = \left( \lim_{x \rightarrow c} f(x) \right) + \left( \lim_{x \rightarrow c} g(x) \right)$
3. **Difference:**  $\lim_{x \rightarrow c} (f(x) - g(x)) = \left( \lim_{x \rightarrow c} f(x) \right) - \left( \lim_{x \rightarrow c} g(x) \right)$
4. **Constant Multiple:**  $\lim_{x \rightarrow c} af(x) = a \lim_{x \rightarrow c} f(x)$
5. **Product:**  $\lim_{x \rightarrow c} (f(x)g(x)) = \left( \lim_{x \rightarrow c} f(x) \right) \left( \lim_{x \rightarrow c} g(x) \right)$
6. **Quotient:**  $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\left( \lim_{x \rightarrow c} f(x) \right)}{\left( \lim_{x \rightarrow c} g(x) \right)}$ , provided  $\lim_{x \rightarrow c} g(x) \neq 0$
7. **Power:**  $\lim_{x \rightarrow c} (f(x))^n = \left( \lim_{x \rightarrow c} f(x) \right)^n$
8. **Fractional Power:**  $\lim_{x \rightarrow c} (f(x))^{\frac{n}{m}} = \left( \lim_{x \rightarrow c} f(x) \right)^{\frac{n}{m}}$ , provided  $f(x) \geq 0$  for  $x$  near  $c$  if  $m$  is even and  $m/n$  is reduced to lowest terms
9. **Polynomial Functions:**  $\lim_{x \rightarrow c} p(x) = p(c)$  if  $p(x)$  is a polynomial

**Example 9.** Find each limit if it exists.

1.  $\lim_{x \rightarrow 2} (7x + 2)$

2.  $\lim_{x \rightarrow -3} (5x)$

3.  $\lim_{x \rightarrow 5} x(4x + 1)$

4.  $\lim_{x \rightarrow -1} \left( \frac{x + 8}{x + 2} \right)$

5.  $\lim_{x \rightarrow -3} \sqrt{-5x + 1}$

**Example 10.** Find the value of the following limits given that

$$\lim_{x \rightarrow 3} f(x) = 4 \text{ and } \lim_{x \rightarrow 3} g(x) = -3.$$

1.  $\lim_{x \rightarrow 3} 7f(x)$

2.  $\lim_{x \rightarrow 3} (3f(x) - 2g(x))$

3.  $\lim_{x \rightarrow 3} \left( \frac{2f(x)}{3g(x)} \right)$

**Example 11.** Let  $f(x) = \begin{cases} x^2 + 2 & \text{if } x < 1 \\ 4x - 1 & \text{if } x \geq 1 \end{cases}$ . Find the following quantities:

1.  $\lim_{x \rightarrow 1^-} f(x)$

2.  $\lim_{x \rightarrow 1^+} f(x)$

3.  $\lim_{x \rightarrow 1} f(x)$

**Example 12.** Let  $f(x) = \begin{cases} 5x - 6 & \text{if } x \leq 2 \\ 2x + 4 & \text{if } x > 2 \end{cases}$ . Find the following quantities:

1.  $\lim_{x \rightarrow 2^-} f(x)$

2.  $\lim_{x \rightarrow 2^+} f(x)$

3.  $\lim_{x \rightarrow 2} f(x)$

**Example 13.** Let  $f(x) = \frac{x^2 + 4x - 5}{x - 1}$ . Find the following quantities:

1.  $\lim_{x \rightarrow 1} f(x)$

2.  $\lim_{x \rightarrow -1} f(x)$

3.  $\lim_{x \rightarrow -5} f(x)$

**Example 14.** Let  $f(x) = \frac{|x - 2|}{x - 2}$ . Find the following quantities:

1.  $\lim_{x \rightarrow 2^-} f(x)$

2.  $\lim_{x \rightarrow 2^+} f(x)$

3.  $\lim_{x \rightarrow 2} f(x)$

**Theorem 2** (The Squeeze Theorem).

Assume the functions  $f$ ,  $g$ , and  $h$  satisfy  $f(x) \leq g(x) \leq h(x)$  for all values of  $x$  near  $c$ , except possibly at  $c$ . If

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} h(x) = L,$$

then  $\lim_{x \rightarrow c} g(x) = L$ .

**Example 15.** Use the squeeze theorem to verify that

$$\lim_{x \rightarrow 0} x^2 \sin(1/x) = 0.$$

**Example 16.** Suppose  $f$  satisfies  $1 \leq f(x) \leq 1 + x^2$  for all values of  $x$  near 0. Find

$$\lim_{x \rightarrow 0} f(x).$$

## SECTION 2.4 - INFINITE LIMITS

Two more limit scenarios are frequently encountered in calculus and are discussed in the next two sections: *infinite limits* and *limits at infinity*. An *infinite limit* occurs when function values increase or decrease without bound near a point. The other type of limit, known as a *limit at infinity*, occurs when the independent variable increases or decreases without bound. The ideas behind infinite limits and limits at infinity are quite different. Therefore, it is important to distinguish these limits and the methods used to calculate them. If one has an *infinite limit*; in general, the dependent variable (i.e. the value of the function) becomes *arbitrarily large* in magnitude as the independent variable approaches a *finite number*. If one has a limit at infinity, the opposite occurs, and we'll learn more about that in the next section.

**Definition 7** (One-Sided Infinite Limits).

Suppose  $f$  is defined for all  $x$  near  $c$  where  $x > c$ . If  $f(x)$  becomes arbitrarily large for all  $x$  sufficiently close to  $c$  with  $x > c$ , we write

$$\lim_{x \rightarrow c^+} f(x) = \infty.$$

The one-sided infinite limits  $\lim_{x \rightarrow c^+} f(x) = -\infty$ ,  $\lim_{x \rightarrow c^-} f(x) = \infty$  and  $\lim_{x \rightarrow c^-} f(x) = -\infty$  are defined analogously.

**Definition 8** (Infinite Limits).

Suppose  $f$  is defined for all  $x$  near  $c$ . If  $f(x)$  grows arbitrarily large for all  $x$  sufficiently close (but not equal) to  $c$ , we write

$$\lim_{x \rightarrow c} f(x) = \infty$$

and say the limit of  $f(x)$  as  $x$  approaches  $c$  is infinity. If  $f(x)$  is negative and grows arbitrarily large in magnitude for all  $x$  sufficiently close (but not equal) to  $c$ , we write

$$\lim_{x \rightarrow c} f(x) = -\infty$$

and say the limit of  $f(x)$  as  $x$  approaches  $c$  is negative infinity. *In both cases, the limit does not exist.*

Both infinite limits and limits at infinity are each associated with a topic you are likely already familiar with: vertical asymptotes and horizontal asymptotes. In the case of infinite limits, we produce a way of characterizing *vertical asymptotes*.

**Definition 9** (Vertical Asymptotes).

Suppose  $f$  is defined for all  $x$  near  $c$ . If

$$\lim_{x \rightarrow c^-} f(x) = \pm\infty, \quad \lim_{x \rightarrow c^+} f(x) = \pm\infty, \quad \text{or} \quad \lim_{x \rightarrow c} f(x) = \pm\infty,$$

we say the line  $x = a$  is a *vertical asymptote* of the function  $f$ .



**Example 17.**

Observe that the function  $\frac{1}{x}$  grows arbitrarily large as  $x$  approaches 0 from the right. This tells us that  $\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$ . Use this fact to analyze the limit

$$\lim_{x \rightarrow 0^+} \frac{x+5}{x}.$$

**Example 18.**

Analyze  $\lim_{x \rightarrow -4^+} \frac{-x^3 + 5x^2 - 6x}{-x^3 - 4x^2}$ .

**Example 19.**

Find the vertical asymptotes of  $f(x) = \frac{x^2 - 4x + 3}{x^2 - 1}$ .

## SECTION 2.5 - LIMITS AT INFINITY

In the last section, we looked at infinite limits – i.e. limits in which the dependent variable increases or decreases without bound as the independent variable approaches a finite number. With limits at infinity, the opposite occurs: The dependent variable approaches a finite number as the independent variable becomes arbitrarily large in magnitude.

**Definition 10** (Limits at Infinity and Horizontal Asymptotes).

If  $f(x)$  becomes arbitrarily close to a finite number  $L$  for all sufficiently large and positive  $x$ , then we write

$$\lim_{x \rightarrow \infty} f(x) = L.$$

In such cases, we say the limit of  $f(x)$  as  $x$  approaches infinity is  $L$ . In this case, the line  $y = L$  is a horizontal asymptote of  $f$ . The limit at negative infinity,  $\lim_{x \rightarrow -\infty} f(x) = M$ , is defined analogously. When this limit exists,  $y = M$  is a horizontal asymptote.

Of course, it can be the case that we have *infinite limits at infinity* as well (for example, think of the function  $f(x) = x^2$ , which grows arbitrarily large as  $x$  grows arbitrarily large). This observation can be made precise with the following definition.

**Definition 11** (Infinite Limits at Infinity).

If  $f(x)$  becomes arbitrarily large as  $x$  becomes arbitrarily large, then we write

$$\lim_{x \rightarrow \infty} f(x) = \infty.$$

The limits  $\lim_{x \rightarrow \infty} f(x) = -\infty$ ,  $\lim_{x \rightarrow -\infty} f(x) = \infty$ , and  $\lim_{x \rightarrow -\infty} f(x) = -\infty$  are defined similarly.

When  $f(x)$  is a polynomial, we have a few nice rules to help us evaluate such limits.

**Theorem 3** (Limits at Infinity of Powers and Polynomials).

Let  $n$  be a positive integer and let  $p$  be the polynomial

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0, \text{ where } a_n \neq 0.$$

Then,

1.  $\lim_{x \rightarrow \pm\infty} x^n = \infty$  when  $n$  is even
2.  $\lim_{x \rightarrow \infty} x^n = \infty$  and  $\lim_{x \rightarrow -\infty} x^n = -\infty$  when  $n$  is odd
3.  $\lim_{x \rightarrow \pm\infty} \frac{1}{x^n} = \lim_{x \rightarrow \pm\infty} x^{-n} = 0$
4.  $\lim_{x \rightarrow \pm\infty} p(x) = \lim_{x \rightarrow \pm\infty} a_n x^n = \pm\infty$  depending on  $n$  (i.e. the degree of the polynomial) and the sign of  $a_n$  (i.e. the leading coefficient).

**Example 20.**

Evaluate the limits as  $x \rightarrow \pm\infty$  of the following functions:

1.  $p(x) = 3x^4 - 6x^2 + x - 10$

3.  $z(x) = 2x^6 - x^2 + 12$

2.  $q(x) = -2x^3 + 3x^2 - 12$

4.  $w(x) = x^2 - \frac{1}{x^2}$

**Example 21.**

Determine the *end behavior* (that is, the limit as  $x \rightarrow \pm\infty$ ) of the following functions:

1.  $f(x) = \frac{3x+2}{x^2-1}$

2.  $g(x) = \frac{40x^4+4x^2-1}{10x^4+8x^2+1}$

In the previous examples, you may have noticed the beginnings of a pattern: the end behavior of a rational function (that is, a function  $f(x) = \frac{p(x)}{q(x)}$ , where  $p(x)$  and  $q(x)$  are polynomials) is determined by the degree of the polynomials in the numerator and denominator and their leading terms. We can make this precise as follows:

**Theorem 4** (End Behavior and Asymptotes of Rational Functions).

Suppose  $f(x) = \frac{p(x)}{q(x)}$  is a rational function, where

$$p(x) = a_mx^m + a_{m-1}x^{m-1} + \dots + a_2x^2 + a_1x + a_0 \text{ and } q(x) = b_nx^n + b_{n-1}x^{n-1} + \dots + b_2x^2 + b_1x + b_0$$

where  $a_m \neq 0$  and  $b_n \neq 0$ . Then,

**1. Degree of numerator less than degree of denominator:**

If  $m < n$ , then  $\lim_{x \rightarrow \pm\infty} f(x) = 0$  and  $y = 0$  is a horizontal asymptote of  $f$ .

**2. Degree of numerator equals degree of denominator:**

If  $m = n$ , then  $\lim_{x \rightarrow \pm\infty} f(x) = a_m/b_n$  and  $y = a_m/b_n$  is a horizontal asymptote of  $f$ .

**3. Degree of numerator greater than degree of denominator:**

If  $m \geq n$ , then  $\lim_{x \rightarrow \pm\infty} f(x) = \infty$  or  $-\infty$  and  $f$  has no horizontal asymptote.

**4. Nonconstant asymptotes:**

If  $m \geq n$ , then while  $f(x)$  doesn't approach a constant value, it does approach the value of the function  $g(x) = p(x) \div q(x)$ , where  $\div$  denotes polynomial long division. In the case where  $m = n + 1$ , then  $f$  approaches a line called a slant asymptote.

**5. Vertical asymptotes:**

Assuming that  $f$  is in reduced form (that is, if  $p$  and  $q$  share no common factors), vertical asymptotes occur at the zeros of  $q$ .

Ocassionally, we might also encounter some special functions – which we call transcendental functions – which are not obviously covered by the case above. The end behavior of a few of these functions is as follows:

**Theorem 5** (End Behavior of  $e^x$ ,  $e^{-x}$ , and  $\ln(x)$ ).

1.  $\lim_{x \rightarrow \infty} e^x = \infty$  and  $\lim_{x \rightarrow -\infty} e^x = 0$

2.  $\lim_{x \rightarrow \infty} e^{-x} = 0$  and  $\lim_{x \rightarrow -\infty} e^{-x} = \infty$

3.  $\lim_{x \rightarrow 0^+} \ln(x) = -\infty$  and  $\lim_{x \rightarrow \infty} \ln(x) = \infty$

**Example 22.**

Determine the end behavior of  $f(x) = \frac{10x^2 + x + 1}{x^{2.2} + \sqrt{x}}$ .

## SECTION 2.6 - CONTINUITY

The graphs of many functions encountered in this text contain no holes, jumps, or breaks. In general, we call functions such as this *continuous*. Informally, we say that a function  $f$  is continuous at  $c$  if the graph of  $f$  does not have a hole or break at  $c$  (that is, if the graph near  $c$  can be drawn without lifting the pencil). If a function is not continuous at  $c$ , then we say that  $c$  is a *point of discontinuity*.

This informal description of continuity is sufficient for determining the continuity of simple functions, but it is not precise enough to deal with more complicated functions. For that, we'll need the following:

**Definition 12** (continuity).

A function  $f$  is *continuous* at a point  $c$  if

$$\lim_{x \rightarrow c} f(x) = f(c).$$

If  $f$  is not continuous at  $c$ , then  $c$  is said to be a *point of discontinuity*.

Think of this definition somewhat like a checklist. Namely, in order for a function  $f$  to be continuous at a point  $c$ , *all* of the following three conditions must hold:

1.  $f(c)$  is defined (i.e.  $c$  is in the *domain* of  $f$ )
2.  $\lim_{x \rightarrow c} f(x)$  exists
3.  $\lim_{x \rightarrow c} f(x) = f(c)$  (i.e. the value of  $f$  at  $c$  equals the limit of  $f(x)$  as  $x$  approaches  $c$ ).

Now, let's look at some useful properties that can allow us to show that some function is continuous by showing that all of the "parts" of the function (in a sense) are continuous.

**Theorem 6** (Continuity Rules).

If  $f$  and  $g$  are continuous at a point  $c$ , then the following functions are also continuous at  $c$ . Assume  $k$  is a constant and  $n > 0$  is an integer.

- |            |                                   |
|------------|-----------------------------------|
| 1. $f + g$ | 4. $fg$                           |
| 2. $f - g$ | 5. $f/g$ , provided $g(c) \neq 0$ |
| 3. $kf$    | 6. $(f(x))^n$                     |

As in previous sections, we also have useful results for polynomials in particular.

**Theorem 7** (Continuity Properties for Polynomial and Rational Functions).

1. A polynomial function is continuous at all values  $x$ .
2. A rational function (recall, that means a function of the form  $\frac{p}{q}$ , where  $p$  and  $q$  are polynomials) is continuous at all values of  $x$  for which  $q(x) \neq 0$ .



Frequently, we will also encounter functions that can be written as a *composition*. In that case, we also have some useful results.

**Theorem 8** (Continuity of Composite Functions at a Point).

If  $g$  is continuous at  $c$  and  $f$  is continuous at  $g(c)$ , then the composite function  $(f \circ g)(x) = f(g(x))$  is continuous at  $c$ .

From the continuity properties of such functions, we can also say that the limit of a composition can be evaluated by direct substitution. That is, we have the following theorem:

**Theorem 9** (Limits of Composite Functions).

1. If  $g$  is continuous at  $c$  and  $f$  is continuous at  $g(c)$ , then

$$\lim_{x \rightarrow c} f(g(x)) = f\left(\lim_{x \rightarrow c} g(x)\right)$$

2. If  $\lim_{x \rightarrow c} g(x) = L$  and  $f$  is continuous at  $L$ , then

$$\lim_{x \rightarrow c} f(g(x)) = f\left(\lim_{x \rightarrow c} g(x)\right)$$

Notice, while the statements in Theorem 9 seem the same, they differ in an important way.

Namely, the first statement requires *both* that  $f$  is continuous at  $g(c)$  and  $g$  is continuous at  $c$ , while the second statement *only* requires that the limit  $\lim_{x \rightarrow c} g(x)$  exists and that  $f$  is continuous at that limiting value. This will let us determine the continuity of functions with roots, as we will now see.

**Theorem 10** (Continuity of Functions with Roots).

Assume that  $m$  and  $n$  are positive integers with no common factors. If  $m$  is an odd integer, then  $(f(x))^{n/m}$  is continuous at all points at which  $f$  is continuous. If  $m$  is even, then  $(f(x))^{n/m}$  is continuous at all points  $c$  at which  $f$  is continuous and  $f(c) > 0$ .

After reading the last theorem, you may be wondering why we need  $f(c) > 0$  in the case where  $m$  is even instead of just  $f(c) \geq 0$ . This has to do with the notion of *continuity at endpoints* and *continuity on an interval*, which we will now define.

**Definition 13** (Continuity at Endpoints and Continuity on an Interval).

A function  $f$  is *continuous from the left* (or *left-continuous*) at a point  $c$  if

$$\lim_{x \rightarrow c^-} f(x) = f(c).$$

Similarly, a function  $f$  is *continuous from the right* (or *right-continuous*) at a point  $c$  if

$$\lim_{x \rightarrow c^+} f(x) = f(c).$$

Given an interval  $I$  without endpoints, we say a function  $f$  is *continuous on*  $I$  if it is continuous at all points in  $I$ . If the interval  $I$  contains its endpoints, continuity on  $I$  requires *left-continuity* or *right-continuity* at the endpoints of  $I$  and continuity at all other points of  $I$ .

**Example 23.**

Using the continuity properties above, determine where each of the functions are continuous.

1.  $f(x) = 3x^3 - 4x^2 + x + 7$

2.  $f(x) = \frac{x^3 + 4x^2 + 3x - 7}{x^2 - 15x + 26}$

3.  $f(x) = \frac{3x - 2}{x^2 + 3}$

4.  $f(x) = \sqrt[3]{x - 10}$

5.  $f(x) = \sqrt{x^2 - 25}$

**Example 24.**

Evaluate the limit  $\lim_{x \rightarrow -1} \sqrt{2x^2 - 1}$ .

**Example 25.**

Determine the intervals of continuity for

$$f(x) = \begin{cases} x^2 + 1 & \text{if } x \leq 0 \\ 3x + 5 & \text{if } x > 0 \end{cases}.$$

Frequently, we will encounter a few common transcendental functions in addition to the functions we've mentioned previously. It is often useful to know the continuity properties of such functions, which we will now summarize.

**Theorem 11** (Continuity of Transcendental Functions).

*The following functions are continuous at all points of their domains.*

| Trigonometric |           | Inverse Trigonometric |                | Exponential | Logarithmic |
|---------------|-----------|-----------------------|----------------|-------------|-------------|
| $\sin(x)$     | $\cos(x)$ | $\sin^{-1}(x)$        | $\cos^{-1}(x)$ | $b^x$       | $\log_b(x)$ |
| $\tan(x)$     | $\cot(x)$ | $\tan^{-1}(x)$        | $\cot^{-1}(x)$ | $e^x$       | $\ln(x)$    |
| $\sec(x)$     | $\csc(x)$ | $\sec^{-1}(x)$        | $\csc^{-1}(x)$ |             |             |

**Example 26.**

Evaluate the limit  $\lim_{x \rightarrow 2} \cos\left(\frac{x^2 - 4}{x - 2}\right)$ .

**Example 27.**

Evaluate the limit  $\lim_{x \rightarrow 0} \frac{\cos^2(x) - 1}{\cos(x) - 1}$ .

A common problem in mathematics is finding solutions to equations of the form  $f(x) = L$ . Before attempting to find values of  $x$  satisfying this equation, it is worthwhile to determine whether a solution exists. The existence of solutions is often established using a result known as the *Intermediate Value Theorem*, which we will now state.

**Theorem 12 (Intermediate Value Theorem).**

Suppose  $f$  is continuous on the interval  $[a, b]$  and  $L$  is a number strictly between  $f(a)$  and  $f(b)$  (that is,  $f(a) < L < f(b)$  or  $f(b) < L < f(a)$ ). Then, there exists at least one number  $c$  in  $(a, b)$  satisfying  $f(c) = L$ .

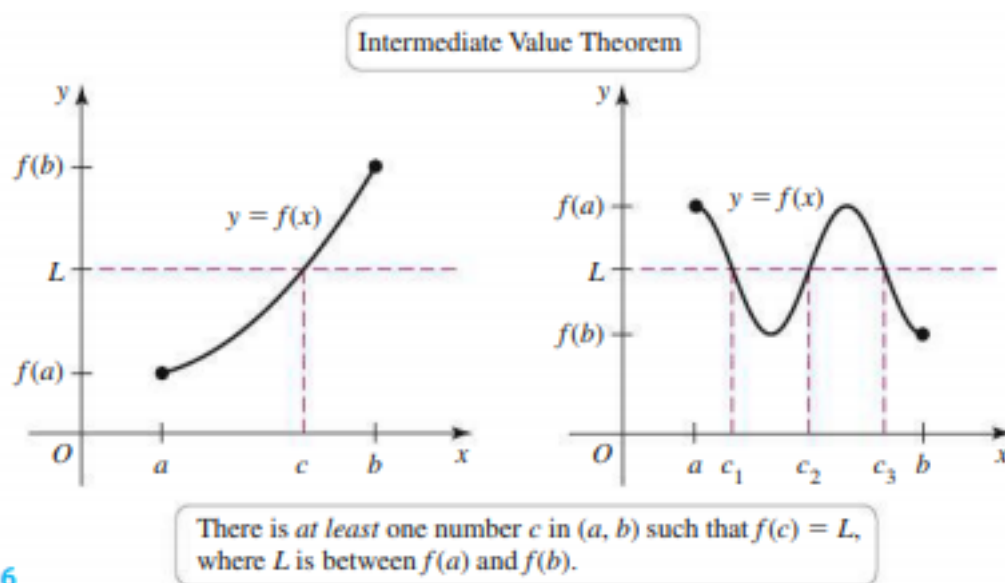


Figure 2.56

**Example 28.**

Suppose you invest \$1000 in a special 5-year savings account with a fixed annual interest rate  $r$ , with monthly compounding. The amount of money  $A$  in the account after 5 years (60 months) is

$$A(r) = 1000 \left( 1 + \frac{r}{12} \right)^{60}.$$

Your goal is to have \$1400 in the account after 5 years. Use the Intermediate Value Theorem to show there is a value of  $r$  in  $(0, 0.08)$  – that is, an interest rate between 0% and 8% – for which  $A(r) = 1400$ .